Smart Gate v3: Electrical Wiring Diagram REV 0.4

1. Purpose

This document is the as-built electrical record for the Smart Gate v3 test bench. It accompanies the two diagram sheets and explains the power domains, every Jetson header connection, the pin configuration fix the motor pins need, safety rules, revision history and open items.

2. Diagram sheets

- Sheet 1: power and actuation (`docs/wiring/sheet1_power_actuation.png`)
- Sheet 2: sensors and Jetson header (`docs/wiring/sheet2_sensors_header.png`)

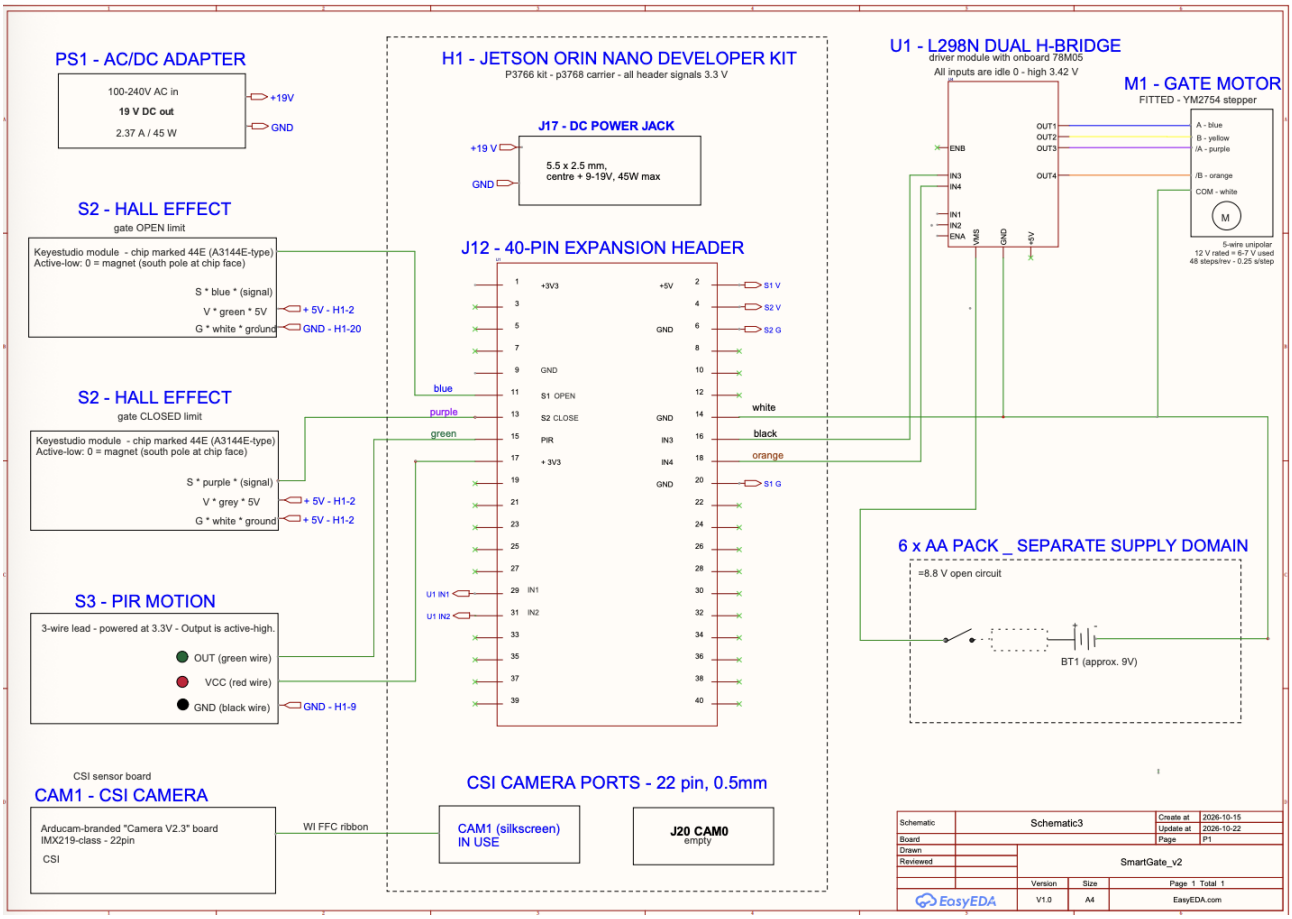


Figure 1: v3 SmartGate Schematic Electrical Diagram

3. Power domains

Domain	Source	Feeds	Notes
Compute	Jetson Orin Nano DC supply	Jetson, camera (via CSI), 3.3 V logic	Kept separate from the motor supply
Motor	6×AA battery pack (about 6–7 V)	L298N motor input, stepper	12 V-rated motor runs under-voltage; no fuse or switch (open item)
Sensor	Jetson 5 V header pin	Hall-effect sensors	Sensors need about 4.5 V or more; output stays at a safe 3.34 V
Ground	Common ground between Jetson (pin 6) and L298N GND	All logic signals	Needed so the Jetson's GPIO levels are read correctly

4. Jetson 40-pin header connections

BOARD (physical) pin numbering, as used in `src/main/io_control.py`.

Header pin	Function	Connected to	Direction
29	Stepper IN1 *(checked against repo)*	L298N IN1 → OUT1 (blue)	Output
31	Stepper IN2 *(checked against repo)*	L298N IN2 → OUT2 (yellow)	Output
16	Stepper IN3 *(checked against repo)*	L298N IN3 → OUT3 (purple)	Output
18	Stepper IN4 *(checked against repo)*	L298N IN4 → OUT4 (orange)	Output
11	Gate open sensor	Hall sensor (open)	Input
13	Gate close sensor	Hall sensor (close)	Input
15	Motion trigger *(pin checked against repo)*	PIR sensor output (VCC, OUT and GND roles inferred from wire colours; see Section 8)	Input
2 (and 4)	Hall sensor and PIR supply	Both Hall sensors and PIR, 5 V	Power
6 (and 9)	Ground	L298N, Hall sensors, PIR	Ground
CSI	Camera	Arducam 8MP V2.3 (IMX219); CAM1 port, IMX219-C overlay; code uses `sensor-id=0`	—

The stepper's white common wire goes to GND, and the L298N's ENA/ENB jumpers are fitted *(from code comments)*.

5. Motor pin configuration fix

The Orin's header pins boot as inputs. Four register writes make the motor pins usable as outputs. `setup/sgate_boot.sh` applies them at start-up *(checked against repo)*, run by `smartgate.service` before the container starts. The settings reset on every restart, so the script must run each boot.

Read them back with:

```
for r in 0x243D020 0x243D010 0x2430068 0x2430070; do sudo busybox devmem $r; done
```

Header pin	Register	Expected readback after fix	Readback if not applied
16	`0x243D020`	5	55
18	`0x243D010`	5	55
29	`0x2430068`	8	58
31	`0x2430070`	8	58

If the readback shows 55, 55, 58, 58, the fix has not taken effect and the motor will not move. Apply it manually with `sudo ~/SmartGate/setup/sgate_boot.sh`.

6. Safety rules

- Never connect the L298N's 5 V output to the Jetson. It is a second power source and would back-feed the board.
- Isolating the motor supply: the pack has no switch; remove one cell.
- Power order: connect the battery pack only after the Jetson is running.
- Before connecting a real gate: fit a fuse and power switch and the Hall resistors (Section 8).

7. Revision history

Revision	Date	Author	Change
REV 0.1	16 Sep 2026	Antus Jojo	Initial diagram redrawn from the inherited v0.2 schematic (DC motor, Jetson Nano) for the Jetson Orin Nano
REV 0.2	15 Oct 2026	Theartha Hari Nair	DC motor replaced by a stepper on pins 29, 31, 16 and 18 (IN1–IN4) after the 15 Oct scope change
REV 0.3	22 Oct 2026	Theartha Hari Nair	Motor-pin configuration fix (Section 5) and PIR on pin 15 added
REV 0.4	1 Nov 2026	Theartha Hari Nair	As built: Jetson Orin Nano, 6×AA pack, IMX219 camera, stepper on pins 16/18/29/31, Hall sensors on pins 11/13 at 5 V

8. Open items

Item	Status
1 kΩ Hall sensor resistors	Planned, not fitted
Fuse and power switch on the motor supply	Not fitted
Motor supply voltage	12 V motor on about 6–7 V; confirm supply for a real gate
PIR pin roles	Inferred from wire colours; confirm against the datasheet
Fan pin	Still listed in `io_control.py` (pin 7) but not wired; causes a harmless "channel 7 is set to input" warning
Second-person check of this diagram	Pin assignments checked against the code (`src/main/io_control.py`) on 1 Nov 2026 by Antus Jojo; physical check of the bench still to do